⑤ User data containing a shell script that is executed during bootstrapping to install a Jenkins server

⑥ Downloads and installs Jenkins

⑦ Starts Jenkins

In case the EC2 instance fails, AWS will not replace the instance automatically under all circumstances. Therefore, you need to create a CloudWatch alarm to trigger the recovery of the virtual machine automatically. A CloudWatch alarm consists of the following:

- A metric that monitors data (health check, CPU usage, and so on)
- A rule defining a threshold based on a statistical function over a period of time
- Actions to trigger if the state of the alarm changes (such as triggering a recovery of an EC2 instance if the state changes to `ALARM`)

An alarm can be in one of the following states:

- `OK`—The threshold hasn't been reached.
- `INSUFFICIENT_DATA`—There isn't enough data to evaluate the alarm.
- `ALARM`—The threshold has been tripped.

Listing 13.2 creates a CloudWatch alarm based on a metric called `StatusCheckFailed_System` (referenced by attribute `MetricName`) to monitor a virtual machine's health and recover it if the underlying host system fails. This metric contains the results of the system status checks performed by the EC2 service every minute. If the check fails, a measurement point with value `1` is added to the metric `StatusCheckFailed_System`. Because the EC2 service publishes this metric, the `Namespace` is called `AWS/EC2` and the `Dimension` of the metric is the ID of a virtual machine.

The CloudWatch alarm checks the metric every 60 seconds, as defined by the `Period` attribute. As defined in `EvaluationPeriods`, the alarm will check the last five periods—the last five minutes in this example. The check runs a statistical function specified in `Statistic` on the time periods. The result of the statistical function, a maximum function in this case, is compared against `Threshold` using the chosen `ComparisonOperator`. If the result is negative, the alarm actions